Fundamentals of Building AI Agents ✕

Today's goal • 3/3 items

Today's Skill Points 101 XP

See skill progress

Module 1
Foundations of Tool Calling and Chaining

Module 2
LCEL and Manual Tool Calling in LangChain

Introduction to Chaining and LCEL Basics
- ✓ LangChain LCEL Chaining Method
  Video • 4 min
- ✓ Lab: AI Powered Data Analysis with LCEL
  Ungraded App Item • 45 min
- ✓ Cheat Sheet: LangChain Expression Language (LCEL)
  Reading • 5 min
- ✓ Practice Quiz: Introduction to Chaining & LCEL Basics
  Practice Assignment • Grade: 100%

Manual Tool Calling Basics
- ✓ When to Call Tools Manually
  Video • 4 min
- ✓ Structured Outputs for Tool Calling
  Reading • 5 min
- ✓ Practice Quiz: Manual Tool Calling Basics
  Practice Assignment • Grade: 100%

Parsing and Validating Tool Calls
- ✓ Build LLM Agents with Tools
  Video • 4 min
- ✓ Build Interactive LLM Agents
  Video • 5 min
- ✓ Lab: Build Interactive LLM Agents with Tools
  Ungraded App Item • 1h
- ✓ Lab: Build a Tool Calling Agent
  Ungraded App Item • 1h
- ✓ Practice Quiz: Parsing and Validating Tool Calls
  Practice Assignment • Grade: 100%

Module Summary and Evaluation
- ✓ Summary and Highlights: Introduction to Chaining and LCEL Basics
  Reading • 2 min
- ✓ Cheat Sheet: Manual Tool Calling in LangChain
  Ungraded Plugin • 20 min
- ✓ Graded Quiz: Manual Tool Calling in LangChain
  Graded Assignment • Grade: 100%

Module 3
Using Built-in Agents in LangChain

Natural Language Data Visualization
- ✓ From Natural Language to Data Visualizations with LangChain
  Video • 7 min
- ✓ Bridging Language and Data: How AI Transforms Analytics
  Reading • 5 min
- ✓ Lab: Build Your Own Data Visualization Agent
  Ungraded App Item • 30 min
- ✓ Practice Quiz: Natural Language Data Visualization
  Practice Assignment • Grade: 100%

Conversational Database Access
- ✓ An Introduction to AI-Powered SQL Agents
  Video • 3 min
- ✓ Implementing LangChain's AI-Powered SQL Agent
  Video • 8 min
- ✓ Natural Language Interfaces for Data Systems
  Reading • 10 min
- ✓ Lab: Build a Natural Language SQL Agent
  Ungraded App Item • 1h
- ✓ Practice Quiz: Conversational Database Access
  Practice Assignment • Grade: 100%

Module Summary and Evaluation
- ○ **Summary and Highlights: Using Built-in Agents in LangChain**
  Reading • 2 min
- ○ Cheat Sheet: Using Built-in Agents in LangChain
  Reading • 20 min
- ○ Graded Quiz: Using Built-in Agents in LangChain
  Graded Assignment • 21 min

Course Wrap-Up
- ○ Course Wrap-Up
  Video • 4 min
- ○ Congratulations and Next Steps
  Reading • 10 min
- ○ Team and Acknowledgement
  Reading • 2 min

Summary and Highlights: Using Built-in Agents in LangChain

4 XP • Generative AI Assistance | 4 XP • BI Operations, Governance, and AI

0:04 / 1:23 1x

Congratulations! You have completed this module. At this point in the course, you know:

- You can analyze and visualize data by asking natural language questions to the Pandas DataFrame agent, which instantly generates code and returns clear data insights.
- The agent-generated Python code directly interacts with your DataFrame, filtering, aggregating, and visualizing data based on your natural language prompts.
- Always use sandboxed environments, design clear prompts, validate LLM analysis with human expertise, and iteratively refine your queries for safe and effective AI-driven data analysis.
- AI-powered SQL agents provide a broader range of users with the ability to access and interpret data without the need for deep technical skills. To build apps that enable natural language processing, set up a development environment with an LLM, SQL database, and LangChain's SQL agent to handle natural language queries.
- To run a query, you use a natural language query along with the SQL agent, which translates the query to SQL and retrieves results from the database.
- To develop applications that use natural language processing, you can begin by creating a Python virtual environment. Then, install the required libraries, including LangChain, install your LLMs, launch the SQL server, and build a database connector for LangChain.

Mark as completed

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